

RF and Microwave Engineering Numerical Companion

EX 716 (BEI) / EX 752 (BEX) · IV-I / IV-II · TU, IOE · Chapter 4, all 8 PYQ waveguide and junction problems, 2069–2082 BS

4. RF/Microwave Components and Devices: Numerical Problems

8 solved PYQ problems · 46 marks across 8 papers · 2 guide-parameter sets, 1 cut-off family, 2 mode rankings, 3 junction networks

READ THIS FIRST

Chapter 4 numerals come in two kinds, and neither needs more than a calculator's square root.

- **Waveguide arithmetic** (Problems 1 to 5). Everything hangs off one number, the **factor** $F = \sqrt{1 - (f_c/f)^2}$. Compute it once, write it in a box, and every other quantity is a multiplication or a division by it:

$$\lambda_g = \frac{\lambda}{F}, \quad v_p = \frac{c}{F}, \quad v_g = cF, \quad Z_{TE} = \frac{\eta_0}{F}, \quad Z_{TM} = \eta_0 F.$$

Always start by naming the dominant mode and checking $f > f_c$; a guide below cut-off has no λ_g or v_p to find.

- **Junction networks** (Problems 6 to 8). Treat each tee as its §3.6 matrix and **follow one unit wave through**: out of one tee, into the next, out again. A matched termination deletes a row and column; a connection hands a leaving wave to the other tee as an arriving one. Column power sums are the self-check.
- Constants used by the papers and here: $c = 3 \times 10^8$ m/s, $\eta_0 = 120\pi = 376.99 \Omega$. Every answer is asserted in `src/wg.py`.

Problem 1: Full parameter set of an air-filled rectangular guide 6

80 Ba · width 3 cm, 6 GHz, dominant mode

QUESTION AS PRINTED 80 Ba [6+4]

For an air-filled rectangular waveguide with a width of 3 cm and a desired frequency of operation of 6 GHz (for dominant mode), determine cut-off frequency, cut-off wavelength, group velocity, phase velocity, propagation wavelength in the waveguide and the characteristic impedance. Explain the four basic parameters used to describe the performance of a directional coupler.

Given

- Air filling: $v = c = 3 \times 10^8$ m/s, $\eta = \eta_0 = 120\pi \Omega$.
- Width (broad wall) $a = 3$ cm = 0.03 m; $f = 6$ GHz.
- Dominant mode of a rectangular guide: **TE₁₀** ($m = 1, n = 0$), so b is not needed.

Step 1: cut-off frequency and wavelength of TE₁₀

- $f_c = \frac{c}{2} \sqrt{\left(\frac{1}{a}\right)^2 + 0} = \frac{c}{2a} = \frac{3 \times 10^8}{2 \times 0.03} = \boxed{5 \text{ GHz}}$.
- $\lambda_c = 2a = \boxed{6 \text{ cm}}$ (check: $c/f_c = 3 \times 10^8 / 5 \times 10^9 = 0.06$ m).
- $f = 6 \text{ GHz} > f_c = 5 \text{ GHz} \Rightarrow$ **the guide propagates**, so the remaining quantities exist.

Step 2: the factor, computed once

- $f_c/f = 5/6 = 0.83333$; $(f_c/f)^2 = 0.69444$; $1 - 0.69444 = 0.30556$.
- $F = \sqrt{0.30556} = \boxed{0.55277}$ (exactly $\sqrt{11}/6$).
- Free-space wavelength $\lambda = c/f = 3 \times 10^8 / 6 \times 10^9 = 5$ cm.

Step 3: every other quantity from F

Quantity	Formula	Substitution	Value
Group velocity	$v_g = cF$	$3 \times 10^8 \times 0.55277$	1.658×10^8 m/s
Phase velocity	$v_p = c/F$	$3 \times 10^8 / 0.55277$	5.427×10^8 m/s
Guide wavelength	$\lambda_g = \lambda/F$	$5/0.55277$	9.045 cm
Impedance (TE)	$Z_{TE} = \eta_0/F$	$376.99/0.55277$	682.0 Ω
Phase constant	$\beta = 2\pi/\lambda_g$	$2\pi/0.090453$	69.46 rad/m

- **Checks:** $v_p v_g = 5.427 \times 10^8 \times 1.658 \times 10^8 = 9.0 \times 10^{16} = c^2 \checkmark$; and $\frac{1}{\lambda^2} = \frac{1}{\lambda_g^2} + \frac{1}{\lambda_c^2}$: $400 = 122.2 + 277.8 \text{ m}^{-2} \checkmark$.
- “Characteristic impedance” of a TE guide is its wave impedance Z_{TE} ; it is above η_0 , as it must be for TE.
- **Rider (4 marks):** coupling, directivity, isolation, insertion loss, with $I = C + D$. See theory §4.6.

Problem 2: Same parameters, with suitably assumed dimensions 6

77 Ch · the choice of dimensions is part of the marks

QUESTION AS PRINTED 77 Ch [6]

For a rectangular waveguide, with suitably assumed breadth, width and frequency, (for dominant mode), determine cut-off frequency, phase velocity, propagation wavelength in the waveguide and the characteristic impedance.

Step 0: choose the data, and justify it

- Take the standard X-band guide **WR-90**: $a = 2.286 \text{ cm}$, $b = 1.016 \text{ cm}$, air filled, at $f = 10 \text{ GHz}$.
- Why it is a good choice: it runs in **single-mode** TE_{10} at 10 GHz. The next modes are cut off:
 - TE_{20} : $f_c = c/a = 3 \times 10^8 / 0.02286 = 13.12 \text{ GHz} > 10 \text{ GHz}$.
 - TE_{01} : $f_c = c/2b = 3 \times 10^8 / 0.02032 = 14.76 \text{ GHz} > 10 \text{ GHz}$.
- Any other data is acceptable if it is stated and above cut-off; the method does not change.

Step 1: cut-off

- $f_c = \frac{c}{2a} = \frac{3 \times 10^8}{2 \times 0.02286} = \boxed{6.562 \text{ GHz}}$, $\lambda_c = 2a = 4.572 \text{ cm}$. $10 > 6.562 \checkmark$ propagates.

Step 2: the factor

- $f_c/f = 0.65617$; squared 0.43056; $1 - 0.43056 = 0.56944$; $F = \sqrt{0.56944} = \boxed{0.75461}$.
- $\lambda = c/f = 3 \text{ cm}$.

Step 3: the asked quantities

Quantity	Formula	Substitution	Value
Phase velocity	$v_p = c/F$	$3 \times 10^8 / 0.75461$	3.976×10^8 m/s
Guide wavelength	$\lambda_g = \lambda/F$	$3/0.75461$	3.976 cm
Impedance (TE)	$Z_{TE} = \eta_0/F$	$376.99/0.75461$	499.6 Ω
Group velocity (extra)	$v_g = cF$	$3 \times 10^8 \times 0.75461$	$2.264 \times 10^8 \text{ m/s}$

- Check: $1/\lambda^2 = 1111.1$ and $1/\lambda_g^2 + 1/\lambda_c^2 = 632.7 + 478.4 = 1111.1 \text{ m}^{-2} \checkmark$.

Problem 3: Does the guide pass the signal? Cut-off checks 3 2

70 Bh, 80 Bh · one method, two instances; 70 Bh worked in full

QUESTION AS PRINTED 70 Bh [7+3]

Define expressions for various field components of a rectangular waveguide in TE mode. Show that a 1 GHz signal cannot propagate in TE_{10} made in a rectangular waveguide with a wall separation of 5 cm.

QUESTION AS PRINTED 80 Bh [1+1+2]

What is dominant mode in a waveguide? What are the dominant modes for rectangular and circular waveguides? Does a rectangular waveguide of width 2.254 cm support the propagation of a signal having frequency 6 GHz?

Method

1. Name the mode (TE₁₀, the dominant one) and read the broad-wall width a .
2. $f_c = c/2a$.
3. Compare: $f > f_c$ propagates; $f < f_c$ is evanescent.
4. For “show that”, add the attenuation: $\gamma = \alpha = \sqrt{k_c^2 - k^2}$ is real.

70 Bh worked: $a = 5$ cm, $f = 1$ GHz

- $f_c = \frac{3 \times 10^8}{2 \times 0.05} = 3 \times 10^9$ Hz = 3 GHz.
- $f = 1$ GHz $< f_c = 3$ GHz $\Rightarrow$ below cut-off.
- Confirm with γ :
 - $k_c = \pi/a = \pi/0.05 = 62.832$ rad/m, $k_c^2 = 3947.8$.
 - $k = 2\pi f/c = 2\pi \times 10^9/3 \times 10^8 = 20.944$ rad/m, $k^2 = 438.6$.
 - $\gamma^2 = k_c^2 - k^2 = 3947.8 - 438.6 = 3509.2 > 0 \Rightarrow \gamma$ is real: $\gamma = \alpha = \sqrt{3509.2} = 59.24$ Np/m, $\beta = 0$.
- The field decays as $e^{-59.24z}$; to $1/e$ in $1/59.24 = 1.69$ cm, i.e. $20 \log_{10} e \times 59.24 = 514.5$ dB/m. **No wave travels: the 1 GHz signal cannot propagate.**

Both instances

Paper	Given	$f_c = c/2a$	Compare	Answer
70 Bh	$a = 5$ cm, 1 GHz	3.00 GHz	$1 < 3$	cannot propagate ; $\alpha = 59.2$ Np/m
80 Bh	$a = 2.254$ cm, 6 GHz	$3 \times 10^8/0.04508 = 6.65$ GHz	$6 < 6.65$	not supported

- **80 Bh theory part:** dominant mode = lowest cut-off mode; rectangular TE₁₀, circular TE₁₁ (§4.2).
- The handwritten local example (Waveguide.pdf p6) runs the same 2.254 cm guide at 1 GHz and reaches the same verdict with $f_c = 6.65$ GHz.

Problem 4: Ranking TM modes when $a = 3b$ 2+2

72 Ma

QUESTION AS PRINTED 72 Ma [6+2+2]

Describe magic Tee based on S-parameters. Differentiate between dominant and degenerate modes. Consider a rectangular waveguide having dimension of $a = 3b$ and find the dominant mode among TM₀₁, TM₁₀, TM₁₁, TM₂₁, TM₁₂, TM₀₂ and TM₂₀.

Step 1: put the cut-off formula in terms of a only

- $f_c = \frac{c}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$, and $b = a/3 \Rightarrow n/b = 3n/a$:

$$f_c = \frac{c}{2a} \sqrt{m^2 + 9n^2}$$

- So each mode is ranked by the number $\sqrt{m^2 + 9n^2}$ (in units of $c/2a$).

Step 2: test every listed mode

Mode	(m, n)	Exists? (TM needs $m, n \geq 1$)	$\sqrt{m^2 + 9n^2}$	Rank
TM ₀₁	(0, 1)	no, $E_z \propto \sin 0 = 0$	n/a	excluded
TM ₁₀	(1, 0)	no	n/a	excluded
TM ₁₁	(1, 1)	yes	$\sqrt{10} = 3.162$	1, dominant
TM ₂₁	(2, 1)	yes	$\sqrt{13} = 3.606$	2
TM ₁₂	(1, 2)	yes	$\sqrt{37} = 6.083$	3
TM ₀₂	(0, 2)	no	n/a	excluded
TM ₂₀	(2, 0)	no	n/a	excluded

- **Answer:** among the listed modes the dominant one is TM₁₁, $f_c = 3.162 c/2a$.
- Worth one line: the guide's overall dominant mode is TE₁₀ at $1 \times c/2a$, which the list does not include.
- Riders: magic tee §4.5; dominant vs degenerate §4.2.

Problem 5: Dominant TE and TM modes of a circular guide, with numbers 2

75 Bh · plus the local worked example as practice

QUESTION AS PRINTED 75 Bh [8+2]

Derive the expression for the field strength for TM waves for a air-filled circular waveguide. Check the dominant mode in TE and TM modes.

Step 1: cut-off in terms of the roots

- TE_{nm}: $f_c = \frac{p'_{nm} c}{2\pi a}$; TM_{nm}: $f_c = \frac{p_{nm} c}{2\pi a}$.
- Same a and c for every mode, so **the smallest root is the dominant mode**.

Step 2: compare the smallest roots

Family	Smallest root	Mode	Verdict
TE	$p'_{11} = 1.841$ (first zero of J'_1)	TE ₁₁	dominant TE mode
TE	$p'_{01} = 3.832$ (first zero of J'_0)	TE ₀₁	higher
TM	$p_{01} = 2.405$ (first zero of J_0)	TM ₀₁	dominant TM mode
TM	$p_{11} = 3.832$ (first zero of J_1)	TM ₁₁	higher

- $1.841 < 2.405 \Rightarrow$ **TE₁₁ is the dominant mode of the guide**, TM₀₁ the dominant TM mode. The derivation part is theory §4.3.

Practice (Waveguide.pdf p13): air-filled, radius 2 cm

Mode	Substitution	f_c	$\lambda_c = 2\pi a/p$
TE ₁₁	$1.8412 \times 3 \times 10^8 / (2\pi \times 0.02)$	4.40 GHz	6.83 cm
TM ₀₁	$2.4048 \times 3 \times 10^8 / (2\pi \times 0.02)$	5.74 GHz	5.23 cm

- Does 3 GHz propagate? $3 < 4.40 \text{ GHz} \Rightarrow$ **no**, not even in the dominant mode.

Problem 6: Two magic tees joined at their E-arms 10

77 Ch · derived by connecting two 4-port matrices; asserted in wg.py

QUESTION AS PRINTED 77 Ch [10]

Explain the properties of two Magic Tees if one connects their E-arms and derive its S-matrix.

Given: the two tees

- Each tee, ports (1, 2) collinear, 3 = E-arm, 4 = H-arm, $r = 1/\sqrt{2}$ (§3.6): $[S_T] = r \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & -1 & 1 \\ 1 & -1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix}$.
- Tee A ports 1, 2, 3, 4; tee B ports 1', 2', 3', 4'. The E-arms 3 and 3' are joined, so they disappear as external ports.

- Joining means: the wave **leaving** arm 3 **enters** arm 3', and vice versa: $a_{3'} = b_3, a_3 = b_{3'}$.
- External ports left: 1, 2, 4 (tee A) and 1', 2', 4' (tee B) $\Rightarrow$ a **6-port**, 6×6 matrix.

Step 1: drive port 1 with a unit wave ($a_1 = 1$, all other inputs zero)

- Inside tee A: $b_3 = S_{31} = r, \quad b_4 = S_{41} = r, \quad b_2 = S_{21} = 0$.
- $b_3 = r$ enters tee B's E-arm: $a_{3'} = r$.
- Inside tee B: $b_{1'} = S_{13} a_{3'} = r \cdot r = \frac{1}{2}, \quad b_{2'} = S_{23} a_{3'} = -r \cdot r = -\frac{1}{2}, \quad b_{4'} = S_{43} a_{3'} = 0$.
- Nothing reflects back into arm 3, because $S_{3'3'} = 0$ (matched E-arm).
- Column for port 1: $(b_1, b_2, b_4, b_{1'}, b_{2'}, b_{4'}) = (0, 0, r, \frac{1}{2}, -\frac{1}{2}, 0)$.
- Power check: $\frac{1}{2} + \frac{1}{4} + \frac{1}{4} = 1 \checkmark$.

Step 2: the other drives, same way

- Port 2: $b_3 = S_{32} = -r, b_4 = r$; tee B receives $-r \Rightarrow b_{1'} = -\frac{1}{2}, b_{2'} = +\frac{1}{2}$.
- Port 4 (H-arm A): $b_1 = b_2 = r$, and $b_3 = S_{34} = 0 \Rightarrow$ **nothing crosses** to tee B.
- Ports 1', 2', 4': mirror images, by symmetry of the connection.

Step 3: the 6-port S-matrix

$$[S] = \begin{matrix} & \begin{matrix} 1 & 2 & 4 & 1' & 2' & 4' \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 4 \\ 1' \\ 2' \\ 4' \end{matrix} & \begin{bmatrix} 0 & 0 & r & \frac{1}{2} & -\frac{1}{2} & 0 \\ 0 & 0 & r & -\frac{1}{2} & \frac{1}{2} & 0 \\ r & r & 0 & 0 & 0 & 0 \\ \frac{1}{2} & -\frac{1}{2} & 0 & 0 & 0 & r \\ -\frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & r \\ 0 & 0 & 0 & r & r & 0 \end{bmatrix} \end{matrix}$$

$r = 1/\sqrt{2}$. Identical to the `connect()` result in `wg.py`.

Step 4: properties

- **Matched**: all six diagonal entries are 0.
- **Reciprocal**: $[S] = [S]^T$.
- **Lossless**: every column's powers sum to 1.
- **Collinear isolation kept**: $S_{12} = S_{1'2'} = 0$.
- **H-arms isolated** from each other and from the far tee: $S_{44'} = 0$.
- **Difference channel crosses**: an antiphase pair at (1, 2) appears entirely at (1', 2'), again antiphase.
- **Sum channel stays**: an in-phase pair at (1, 2) leaves entirely by 4.
- Use: a sum/difference separator, e.g. two monopulse channels feeding a common difference path.

Problem 7: Hybrid tee with a terminated arm 4+4+2

82 Ba · 70 Ma's "hybrid tee" [8] is answered by §4.5 plus this

QUESTION AS PRINTED 82 Ba [4+4+2]

Using proper S-Matrices, mention the properties of a Hybrid-Tee in the situation of (a) terminated E-arm, (b) terminated H-arm and (c) terminated co-planar arms.

Step 0: what "terminated" does to a matrix

- Terminated = a **matched load**, $\Gamma = 0$: whatever leaves that arm is absorbed, nothing comes back, so $a_k = 0$.
- Therefore row k and column k simply **drop out**; the rest of the matrix is unchanged.
- Start from the magic tee, ports 1, 2 collinear (the co-planar arms), 3 = E, 4 = H.
- If the paper meant a **short**, $\Gamma = -1$, the answer is different; that is the 82 Bh case, §3.6.

Case	Reduced S-matrix	Properties
(a) E-arm terminated, ports (1, 2, 4)	$\begin{bmatrix} 0 & 0 & r \\ 0 & 0 & r \\ r & r & 0 \end{bmatrix}$	Matched and reciprocal. From the H-arm: equal, in-phase halves at 1 and 2, no loss. Ports 1 and 2 isolated . From port 1: only half reaches port 4, the other half is absorbed in the E-arm load, so the 3-port is not lossless . An in-phase power divider.
(b) H-arm terminated, ports (1, 2, 3)	$\begin{bmatrix} 0 & 0 & r \\ 0 & 0 & -r \\ r & -r & 0 \end{bmatrix}$	Matched and reciprocal. From the E-arm: equal, antiphase halves at 1 and 2. Ports 1 and 2 isolated. From port 1: half to port 3, half lost in the H-arm load. An antiphase (180°) power divider.
(c) Both co-planar arms terminated, ports (3, 4)	$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$	E and H arms both matched and completely isolated . Any power entering either side arm splits into the two loads and nothing comes out anywhere: the tee acts as a matched termination for both.

- Why (a) and (b) **must** be lossy: they are matched, reciprocal 3-ports, and §3.5 proves no such 3-port can be lossless. The load is where the missing half goes.
- Column check, (a): column 1 = $0 + 0 + \frac{1}{2} = \frac{1}{2}$ (half absorbed); column 4 = $\frac{1}{2} + \frac{1}{2} = 1$ (lossless from the H-arm).

Problem 8: Two magic tees connected in the H-plane 5

78 Ch · a short note; the Problem 6 method with the H-arms joined

QUESTION AS PRINTED 78 Ch [5]

Write short notes on: Combinational effect of two magic tees connected in H-plane.

The paper's typos ("Combinational", "magic tees") are reproduced as printed; read "combinational effect of two magic tees".

Step 1: join H-arm 4 to H-arm 4' ($a_{4'} = b_4$, $a_4 = b_{4'}$) and follow port 1

- Tee A: $b_3 = S_{31} = r$, $b_4 = S_{41} = r$.
- $b_4 = r$ enters tee B's H-arm: $b_{1'} = S_{14} r = \frac{1}{2}$, $b_{2'} = S_{24} r = +\frac{1}{2}$ (in phase), $b_{3'} = S_{34} r = 0$.
- From port 2: $b_4 = S_{42} = +r$ as well, so tee B again gets $+\frac{1}{2}, +\frac{1}{2}$.

Step 2: the 6-port S-matrix

$$[S] = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 1' & 2' & 3' \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 1' \\ 2' \\ 3' \end{matrix} & \begin{bmatrix} 0 & 0 & r & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & -r & \frac{1}{2} & \frac{1}{2} & 0 \\ r & -r & 0 & 0 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & r \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & -r \\ 0 & 0 & 0 & r & -r & 0 \end{bmatrix} \end{matrix}$$

Short-note points

- Matched at all six ports, reciprocal, lossless.
- The **sum** (in-phase) signal crosses between the tees through the joined H-arms and re-appears **in phase** at 1', 2'.
- The **difference** signal stays in its own tee's E-arm.
- Collinear ports of each tee stay isolated; the two E-arms are isolated.
- The exact mirror of Problem 6: E-joining passes the difference, H-joining passes the sum.

Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

Short	Means	Short	Means
RF	radio frequency	MW, M/W	microwave
EM, EMR	electromagnetic (radiation)	TEM	transverse electromagnetic mode
TE	transverse electric mode	TM	transverse magnetic mode
WG	waveguide	TL	transmission line
VSWR	voltage standing wave ratio	SWR	standing wave ratio
VNA	vector network analyzer	SA	spectrum analyzer
DUT	device under test	S-params	scattering parameters
RL	return loss	IL	insertion loss
ML	mismatch loss	NF	noise figure
BW	bandwidth	SNR	signal to noise ratio
Tx, TX	transmitter	Rx, RX	receiver
BTS	base transceiver station	GSM	global system for mobile communications
EIRP	effective isotropic radiated power	ERP	effective radiated power
SAR	specific absorption rate	SARPs	standards and recommended practices
HERP	hazards of EM radiation to personnel	RADAR	radio detection and ranging
ASR	airport surveillance radar	CW	continuous wave

Short	Means	Short	Means
TWT, TWTA	traveling wave tube (amplifier)	BWO	backward wave oscillator
CFA	cross-field amplifier	LNA	low noise amplifier
HPA	high power amplifier	SWS	slow wave structure
O-type	linear beam tube (axial B -field)	M-type	crossed-field tube ($E \perp B$)
MASER	microwave amplification by stimulated emission of radiation	LASER	light amplification by stimulated emission of radiation
MESFET	metal-semiconductor FET	HEMT	high electron mobility transistor
GaAs	gallium arsenide	FET	field effect transistor
BJT	bipolar junction transistor	PIN	p-type, intrinsic, n-type diode
IMPATT	impact ionization avalanche transit-time	TRAPATT	trapped plasma avalanche triggered transit
LPF	low pass filter	HPF	high pass filter
BPF	band pass filter	BSF	band stop filter
MAG	maximum available gain	MSG	maximum stable gain
MIC	microwave integrated circuit	MMIC	monolithic microwave integrated circuit
ADS	advanced design system	IF	intermediate frequency
LO	local oscillator	RFIC	radio frequency integrated circuit
PCB	printed circuit board	SMA	subminiature version A connector

Short	Means	Short	Means
attr(s)	attribute(s)	no.	number
freq	frequency	info	information
ops	operations	config	configuration
rm	remove	vs, Vs	versus
w/, w/o	with, without	Eg	example / for example
+Fig	draw a figure	(I)	importance / need / significance
→	leads to / therefore	+	and
=	is / equals	norm	normalized
max, min	maximum, minimum	calc	calculation / calculate
eq, eqn	equation	approx	approximately
pts	points	coeff	coefficient
temp	temperature	params	parameters

Standard and component terms (proper names and symbols, kept as standard): **Magic Tee** 180° hybrid junction (E-plane & H-plane tee combined) · **Directional Coupler** 4-port forward/reverse coupled wave device · **Circulator** non-reciprocal multi-port ferrite device ($1 \rightarrow 2 \rightarrow 3 \rightarrow 1$) · **Gunn Diode** transferred electron device (GaAs / InP, negative resistance) · **Bolometer** power detector (barretter: PTC, thermistor: NTC) · **Calorimeter** power measurement via heat transfer (static, circulating, dry, flow).

Key mathematical symbols: Z_0 characteristic impedance · Y_0 characteristic admittance · Γ reflection coefficient · S scattering matrix · K Rollett's stability factor · Δ determinant of S-matrix · μ single parameter stability metric · G_T transducer gain · G_{TU} unilateral transducer gain · f_c cutoff frequency · λ_g guide wavelength · v_p phase velocity · v_g group velocity.

Exam-paper month codes (used in the year tags): Ba = Baishakh · Bh = Bhadra · Ash = Ashwin · Ma = Magh · Ch = Chaitra.

Bold year tags are Regular exams, plain are Back exams.

Monospace year tags are BEI (EX 716, IV/I); normal font tags are BEX (EX 752 / EG 785EX, IV/II).

Tier chips over 24 papers, 2069–2082 BS: **TOP /24** = topic asked in 7 or more papers · **HOT /24** = 3 to 6 papers · **PIN /24** = 1 or 2 papers but worth ≥ 6 marks.

Mark chips such as **2+5** are the marks that question actually carried. **syllabus, no PYQ yet** marks a syllabus topic that has never been examined but is written up anyway. **verified/added** marks anything researched and added that was not in the source notes.